📊 INSTRUCTOR-LED COURSE SYLLABUS

Python Data Science

An instructor-led course in 21 three-hour sessions. Learn to turn raw data into answers you can defend: NumPy, pandas, Matplotlib, statistics, SQL, web data and dashboards, all practiced on one fictional company's data and finished with a capstone report.

Course site: rays-python-datascience.netlify.app

`21 sessions × 3 h` `63 class hours` `41 lessons` `2–3 h homework per session` `Python 3.12+`

📚 COURSE DESCRIPTION

Data science is a way of working: ask a clear question, get the data, clean it, explore it, test what you think you see, and show what you found. This course teaches the Python tools for each of those steps in the order you would use them on a real project.

Every session uses the same dataset: two years of records from **Pinecrest Outfitters**, a fictional outdoor-gear retailer with an online shop and four stores. Orders, customers, products, daily store traffic and weather, a customer survey, a checkout-page experiment, a SQLite database, nested website sessions and a deliberately messy export. The web sessions use a practice shop and a practice API built for the course, so your code never points at a real site while you learn.

🎯 By the end you will be able to

- Compute with arrays in NumPy instead of writing loops
- Load, clean, reshape, combine and summarize tables with pandas
- Build clear charts with Matplotlib and seaborn, and interactive ones with Plotly and Bokeh
- Describe data honestly and run a hypothesis test or an A/B test without fooling yourself
- Query a database with SQL from Python, including joins, window functions and indexes
- Collect data from web pages and APIs politely and safely
- Turn an analysis into a small dashboard, and write up findings in a one-page report

✅ PREREQUISITES

- Comfortable writing small Python programs: variables, `if`, loops, functions, lists and dictionaries, reading and writing files, and basic classes, as taught in **Introduction to Programming with Python** and **Intermediate Python** (free at Ray's House of Fun).
- No statistics or prior data experience is needed.
- A Windows, macOS or Linux laptop on which you can install software, with about 2 GB free. Setup is done at home before Session 1 (it follows Lesson 0.1).

💡 FORMAT

21 sessions × 3 hours = 63 class hours, two lessons per session (the capstone session has one). Every lesson on the course site is written for a solo learner and takes 75 to 110 minutes, so two lessons do not fit whole in one room session. The room protects what is hardest alone: the **mental model**, a **live-coding demo** you type along with, a **guided lab** with

help on hand, and **starting** each lesson project. At home you finish the two projects and do each lesson's exercises, quiz and journal entry on the free course site: about **two to three hours** between sessions.

Each session runs 0:00 to 3:00: a warm-up on last session, the first lesson, a 10-minute break, the second lesson, and a wrap-up with an exit ticket and the next bring list.

Twice weekly: about 11 weeks. Two homework sets a week, so plan on four to six hours outside class.

Weekly: 21 weeks, with a full week for each homework set.

Intensive days: two sessions a day; homework becomes evening practice and the capstone is finished afterward.

📖 SESSION BY SESSION

Lesson titles link to the lesson pages on the course site. Homework is the time to finish that session's two lesson projects, exercises, quizzes and journal entries.

SESSION	LESSONS	WHAT YOU WILL BE ABLE TO DO	HOMEWORK
1 **Toolkit and First Arrays** NUMPY	0.1 Your Data Science Toolkit 1.1 Arrays vs Lists	Set up a reproducible Python environment with the course data, load and summarize a real file, read a traceback from the bottom, and create, index and copy NumPy arrays, explaining why they beat lists for numbers.	2–3 h
2 **Vectorized Operations and Broadcasting** NUMPY	1.2 Vectorized Operations 1.3 Broadcasting	Replace loops with whole-array arithmetic, comparisons, boolean masks and aggregations, and combine arrays of different shapes with broadcasting, reading a shape error when it fails.	2–3 h
3 **Randomness and Linear Algebra** NUMPY	1.4 Random Number Generation 1.5 Linear Algebra Essentials	Simulate and sample with a seeded random generator, and use vectors, matrices and numpy.linalg to solve linear systems and fit a first least-squares model.	2–3 h
4 **DataFrames and Loading Data** PANDAS	2.1 DataFrames and Series 2.2 Loading Data	Create, select, filter and extend DataFrames and Series, and load CSV, Excel, JSON and SQL data with the right types from the start.	2–3 h
5 **Cleaning and Grouping** PANDAS	2.3 Cleaning Data 2.4 GroupBy Operations	Repair a deliberately messy export (missing values, duplicates, inconsistent labels, bad dates, text-as-numbers) in a rerunnable pipeline, and summarize by group with split-apply-combine.	2–3 h

SESSION	LESSONS	WHAT YOU WILL BE ABLE TO DO	HOMEWORK
6 **Merging and Reshaping** PANDAS	2.5 Merging and Joining 2.6 Pivot Tables and Reshaping	Combine tables on keys with the right join while accounting for every unmatched row, and move between long and wide data with pivot, melt, stack and crosstab.	2–3 h
7 **Time Series and Windows** PANDAS	2.7 Time Series 2.8 Window Functions	Work with datetime indexes, resampling and gaps in daily data, and compute rolling, expanding and exponentially weighted values, ranks and shifts.	2–3 h
8 **Features and First Charts** PANDAS VISUALIZATION	2.9 Feature Engineering Basics 3.1 Line, Scatter and Bar Charts	Build a leak-free feature table from raw columns, and draw line, scatter and bar charts with Matplotlib's object-oriented interface, choosing the right one for the question.	2–3 h
9 **Composing and Customizing Figures** VISUALIZATION	3.2 Subplots and Composition 3.3 Customizing Plots	Lay out several axes as one readable figure, and use titles, labels, ticks, color and annotation so a chart says one thing clearly.	2–3 h
10 **Saving Figures; Center and Spread** VISUALIZATION STATISTICS	3.4 Saving Figures 4.1 Descriptive Statistics I: Center and Spread	Export figures at the right size, resolution and format for print and the web, and describe data with mean, median, variance and standard deviation, knowing when each misleads.	2–3 h
11 **Shape, Outliers and Distributions** STATISTICS	4.2 Descriptive Statistics II: Shape, Position and Outliers 4.3 Probability Distributions	Describe skew, percentiles and z-scores, investigate outliers by group, and model data with normal, binomial and Poisson distributions and sampling distributions in scipy.stats.	2–3 h
12 **Correlation and Hypothesis Tests** STATISTICS	4.4 Correlation Analysis 4.5 Hypothesis Testing	Measure association with Pearson and Spearman correlation and spot a confound, then test claims with t-tests and chi-square, reporting p-values with effect sizes and assumptions.	2–3 h
13 **A/B Testing and the EDA Workflow** STATISTICS	4.6 A/B Testing 4.7 An EDA Workflow	Size, run and analyze an A/B test without peeking, and explore an unfamiliar dataset with a repeatable routine that ends in written findings. Includes Checkpoint 1.	3.5 h

SESSION	LESSONS	WHAT YOU WILL BE ABLE TO DO	HOMEWORK
14 **SQL from Python: Queries and Joins** SQL	5.1 Database Connections 5.2 Complex Queries	Connect to SQLite with sqlite3 and SQLAlchemy, run safe parameterized queries into pandas, and write joins, subqueries, CTEs (including a recursive one) and CASE expressions.	3 h
15 **SQL Windows and Query Speed** SQL	5.3 SQL Window Functions 5.4 Query Optimization	Compute running totals, rankings and period-over-period changes with SQL window functions, and read a query plan, add an index and measure the difference.	3 h
16 **Documents and HTTP** SQL WEB DATA	5.5 NoSQL with MongoDB 6.1 HTTP with Requests	Query nested documents with MongoDB aggregation pipelines, and make HTTP requests with status codes, headers, timeouts and robots.txt checks.	2–3 h
17 **Scraping Static and Dynamic Pages** WEB DATA	6.2 Parsing HTML with Beautiful Soup 6.3 Dynamic Pages with Selenium	Parse and paginate HTML with Beautiful Soup into a DataFrame, and drive a headless browser with Selenium for content that JavaScript loads, knowing when you do not need one.	2–3 h
18 **APIs: Keys, Tokens and Polite Clients** WEB DATA	6.4 API Authentication 6.5 Rate Limiting and Retries	Call a protected API with keys and expiring bearer tokens kept out of your code, and respect rate limits with retries, backoff and caching. Includes Checkpoint 2.	2–3 h
19 **Seaborn and Plotly** DASHBOARDS	7.1 Statistical Graphics with Seaborn 7.2 Interactive Charts with Plotly	Show distributions, relationships and categories with seaborn, and build interactive Plotly charts with hover, zoom and filtering, saved as HTML or images.	2–3 h
20 **Bokeh and Streamlit** DASHBOARDS CAPSTONE	7.3 Interactive Charts with Bokeh 7.4 Dashboards with Streamlit	Link interactive plots and widgets with Bokeh, turn an analysis into a small Streamlit dashboard, and choose your capstone question with a one-paragraph plan.	3–4 h
21 **Capstone Studio and Showcase** CAPSTONE	8.1 The Pinecrest Report	Carry one business question from plan to a tested, quantified answer in a notebook that reruns, one or two figures, and a one-page report, then present it.	3–5 h

Total out-of-class time: about 48 hours or more across the course, plus the capstone report.

🕑 SESSION OUTLINES

Times run from the start of each session (0:00 to 3:00). Shift the block to any start time.

SESSION 1 Toolkit and First Arrays

You build: 0.1 A project folder with a working environment, plus first_look.py : a short sales summary and a chart; **1.1** A list-versus-array benchmark report on two years of Pinecrest store revenue

0:00–0:30	Welcome, environment check and diagnostic
0:30–1:25	`0.1` Your Data Science Toolkit frame and mental model, live demo, guided lab, start the lesson project
1:25–1:35	*Break*
1:35–2:50	`1.1` Arrays vs Lists frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 2 Vectorized Operations and Broadcasting

You build: 1.2 A store scorecard: conversion rate, basket value and sale-weekend lift for all four Pinecrest stores, with no loop over the 2,903 rows; **1.3** A weekday index for the four Pinecrest stores that removes store size, plus a list of the days that were unusual for their own store

0:00–0:10	Warm-up
0:10–1:25	`1.2` Vectorized Operations frame and mental model, live demo, guided lab, start the lesson project
1:25–1:35	*Break*
1:35–2:50	`1.3` Broadcasting frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 3 Randomness and Linear Algebra

You build: 1.4 A simulation of 10,000 possible months of Boise revenue, a chart of the results, and an honest check of the simulation against the real months; **1.5** A first revenue model for the Asheville store: fitted by least squares, scored, used to predict three scenarios, and checked for its biggest misses

0:00–0:10	Warm-up
0:10–1:20	`1.4` Random Number Generation frame and mental model, live demo, guided lab, start the lesson project
1:20–1:30	*Break*
1:30–2:50	`1.5` Linear Algebra Essentials frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 4 DataFrames and Loading Data

You build: 2.1 A catalog health check for Pinecrest's 242 products; **2.2** A typed Parquet snapshot of seven Pinecrest tables loaded from CSV, Excel and JSON

0:00–0:10	Warm-up
0:10–1:25	2.1 DataFrames and Series frame and mental model, live demo, guided lab, start the lesson project
1:25–1:35	*Break*
1:35–2:50	2.2 Loading Data frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 5 Cleaning and Grouping

You build: 2.3 A rerunnable clean_orders() pipeline with a validation step and a scorecard against the answer key; **2.4** A store scorecard and a customer summary table from two years of orders

0:00–0:10	Warm-up
0:10–1:30	2.3 Cleaning Data frame and mental model, live demo, guided lab, start the lesson project
1:30–1:40	*Break*
1:40–2:50	2.4 GroupBy Operations frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 6 Merging and Reshaping

You build: 2.5 A revenue-by-category report from three merged tables, with every unmatched row accounted for; **2.6** A category-by-month seasonality report for Pinecrest, saved as a tidy CSV

0:00–0:10	Warm-up
0:10–1:25	2.5 Merging and Joining frame and mental model, live demo, guided lab, start the lesson project
1:25–1:35	*Break*
1:35–2:50	2.6 Pivot Tables and Reshaping frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 7 Time Series and Windows

You build: 2.7 A store trend report: clean daily calendar, monthly totals and year-over-year growth for each Pinecrest store; **2.8** A sale-day detector that flags unusual store-days against a leak-free baseline and scores itself against the promotion calendar

0:00–0:10	Warm-up
0:10–1:25	2.7 Time Series frame and mental model, live demo, guided lab, start the lesson project
1:25–1:35	*Break*
1:35–2:50	2.8 Window Functions frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 8 Features and First Charts

You build: 2.9 A leak-free customer feature table with a "will they order again?" target, saved as customer_features.csv; **3.1** A four-chart answer to one question: does rain keep shoppers out of Pinecrest's stores?

0:00–0:10	Warm-up
0:10–1:25	2.9 Feature Engineering Basics frame and mental model, live demo, guided lab, start the lesson project
1:25–1:35	*Break*
1:35–2:50	3.1 Line, Scatter and Bar Charts frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 9 Composing and Customizing Figures

You build: 3.2 A nine-panel figure showing when each product category has its season; **3.3** A presentation-ready chart of online return rates by product category

0:00–0:10	Warm-up
0:10–1:25	3.2 Subplots and Composition frame and mental model, live demo, guided lab, start the lesson project
1:25–1:35	*Break*
1:35–2:50	3.3 Customizing Plots frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 10 Saving Figures; Center and Spread

You build: 3.4 A chart pack: one styled revenue chart per store, saved for the web and for print, plus a single multi-page PDF; **4.1** A "typical order" memo: a summary table for three very different columns, a chart, and a recommendation on which numbers to report

0:00–0:10	Warm-up
0:10–1:15	3.4 Saving Figures frame and mental model, live demo, guided lab, start the lesson project
1:15–1:25	*Break*
1:25–2:50	4.1 Descriptive Statistics I: Center and Spread frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 11 Shape, Outliers and Distributions

You build: 4.2 A store profile sheet: grouped summaries, flagged days investigated, and a box plot comparing the four stores; **4.3** A returns forecast: a checked Poisson model of daily returns, with capacity numbers and a confidence interval

0:00–0:10	Warm-up
0:10–1:15	4.2 Descriptive Statistics II: Shape, Position and Outliers frame and mental model, live demo, guided lab, start the lesson project
1:15–1:25	*Break*
1:25–2:50	4.3 Probability Distributions frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 12 Correlation and Hypothesis Tests

You build: 4.4 A "rain check" analysis: does weather move store traffic, once you compare like with like?; **4.5** A tested, caveated answer to "does rain keep shoppers out of Pinecrest stores?"

0:00–0:10	Warm-up
0:10–1:10	4.4 Correlation Analysis frame and mental model, live demo, guided lab, start the lesson project
1:10–1:20	*Break*
1:20–2:50	4.5 Hypothesis Testing frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 13 A/B Testing and the EDA Workflow

You build: 4.6 A one-page decision memo for the checkout experiment, backed by a notebook that reproduces every number in it; **4.7** A complete exploratory analysis of why Pinecrest orders are returned, ending in a one-page findings note

> Checkpoint 1 (30 minutes): an open-notebook analysis task on Modules 1 to 4.

0:00–0:10	Warm-up
0:10–1:10	4.6 A/B Testing frame and mental model, live demo, guided lab, start the lesson project
1:10–1:20	*Break*
1:20–2:15	4.7 An EDA Workflow frame and mental model, live demo, guided lab, start the lesson project
2:15–2:45	**Checkpoint 1**
2:45–2:50	**Checkpoint hand-in and debrief**
2:50–3:00	Wrap-up

SESSION 14 SQL from Python: Queries and Joins

You build: 5.1 A store scorecard: one reusable, parameterized report function that reads from the database and saves its result back as a new table; **5.2** A returns investigation that joins four tables to find which online orders come back, and why

0:00–0:10	Warm-up
0:10–1:25	5.1 Database Connections frame and mental model, live demo, guided lab, start the lesson project
1:25–1:35	*Break*
1:35–2:50	5.2 Complex Queries frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 15 SQL Windows and Query Speed

You build: 5.3 A monthly store performance table with month-over-month change, year-to-date totals, moving averages and ranks, checked against pandas; **5.4** A tuning report for a three-query workload: plans and timings before and after, the indexes you chose, and what they cost

0:00–0:10	Warm-up
0:10–1:25	5.3 SQL Window Functions frame and mental model, live demo, guided lab, start the lesson project
1:25–1:35	*Break*
1:35–2:50	5.4 Query Optimization frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 16 Documents and HTTP

You build: 5.5 A website funnel report: sessions to cart to checkout to order, overall and by device, with a chart; **6.1** A polite snapshot tool that checks the rules, downloads four pages of the practice shop and logs every request

0:00–0:10	Warm-up
0:10–1:20	5.5 NoSQL with MongoDB frame and mental model, live demo, guided lab, start the lesson project
1:20–1:30	*Break*
1:30–2:50	6.1 HTTP with Requests ground rules and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 17 Scraping Static and Dynamic Pages

You build: 6.2 A category price report scraped from the nine category pages of the practice shop; **6.3** A deals collector that drives a browser through the practice shop's deals page and checks itself against the page's own data file

0:00–0:10	Warm-up
0:10–0:12	Ground rules opener
0:12–1:25	6.2 Parsing HTML with Beautiful Soup frame and mental model, live demo, guided lab, start the lesson project
1:25–1:35	*Break*
1:35–2:50	6.3 Dynamic Pages with Selenium frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 18 APIs: Keys, Tokens and Polite Clients

You build: 6.4 A December returns brief pulled from the protected orders endpoint with your own API client; **6.5** A resumable, logged download of two months of orders that survives an interruption and stays inside the rate limit

> Checkpoint 2 (15 minutes): a short data-access task on Modules 5 and 6.

0:00–0:10	Warm-up
0:10–0:12	Ground rules opener
0:12–1:15	`6.4` API Authentication frame and mental model, live demo, guided lab, start the lesson project
1:15–1:25	*Break*
1:25–2:25	`6.5` Rate Limiting and Retries frame and mental model, live demo, guided lab, start the lesson project
2:25–2:40	**Checkpoint 2**
2:40–2:50	**Checkpoint debrief and Module 6 recap**
2:50–3:00	Wrap-up

SESSION 19 Seaborn and Plotly

You build: 7.1 A four-panel "store performance brief" saved as one PNG; **7.2** sales_explorer.html , a one-file interactive report with three linked-by-topic charts

0:00–0:10	Warm-up
0:10–1:25	`7.1` Statistical Graphics with Seaborn frame and mental model, live demo, guided lab, start the lesson project
1:25–1:35	*Break*
1:35–2:50	`7.2` Interactive Charts with Plotly frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 20 Bokeh and Streamlit

You build: 7.3 store_explorer.html , three linked charts with a store selector and a live selection summary; **7.4** dashboard.py , a Pinecrest store sales dashboard with filters, four metrics, two charts and a CSV download

> Capstone kickoff: choose one of the four Pinecrest questions and leave with a written plan.

0:00–0:10	Warm-up
0:10–1:15	7.3 Interactive Charts with Bokeh frame and mental model, live demo, guided lab, start the lesson project
1:15–1:25	*Break*
1:25–2:30	7.4 Dashboards with Streamlit frame and mental model, live demo, guided lab, start the lesson project
2:30–2:50	**Capstone kickoff**
2:50–3:00	Wrap-up

SESSION 21 Capstone Studio and Showcase

You build: 8.1 A notebook that runs from top to bottom, one or two saved figures, and a one-page findings report (a dashboard is optional)

> Studio with milestone check-ins, then a showcase of every learner's finding.

0:00–0:10	Warm-up
0:10–1:05	8.1 The Pinecrest Report frame: the rubric and the strong sentence, studio 1 · Milestone 5: test and quantify
1:05–1:15	*Break*
1:15–2:40	8.1 The Pinecrest Report studio 2 · Milestone 6: visualize, then the Answer, showcase prep, showcase
2:40–2:50	Course close and what next
2:50–3:00	Wrap-up

📅 SAMPLE SCHEDULES

No dates: fit them to your calendar. "Twice weekly" puts two sessions in each week (Day 1 and Day 2), finishing in week 11.

SESSION	TWICE WEEKLY (≈11 WEEKS)	WEEKLY (21 WEEKS)
1 Toolkit and First Arrays	Week 1, Day 1	Week 1
2 Vectorized Operations and Broadcasting	Week 1, Day 2	Week 2
3 Randomness and Linear Algebra	Week 2, Day 1	Week 3
4 DataFrames and Loading Data	Week 2, Day 2	Week 4

SESSION	TWICE WEEKLY (≈11 WEEKS)	WEEKLY (21 WEEKS)
5 Cleaning and Grouping	Week 3, Day 1	Week 5
6 Merging and Reshaping	Week 3, Day 2	Week 6
7 Time Series and Windows	Week 4, Day 1	Week 7
8 Features and First Charts	Week 4, Day 2	Week 8
9 Composing and Customizing Figures	Week 5, Day 1	Week 9
10 Saving Figures; Center and Spread	Week 5, Day 2	Week 10
11 Shape, Outliers and Distributions	Week 6, Day 1	Week 11
12 Correlation and Hypothesis Tests	Week 6, Day 2	Week 12
13 A/B Testing and the EDA Workflow	Week 7, Day 1	Week 13
14 SQL from Python: Queries and Joins	Week 7, Day 2	Week 14
15 SQL Windows and Query Speed	Week 8, Day 1	Week 15
16 Documents and HTTP	Week 8, Day 2	Week 16
17 Scraping Static and Dynamic Pages	Week 9, Day 1	Week 17
18 APIs: Keys, Tokens and Polite Clients	Week 9, Day 2	Week 18
19 Seaborn and Plotly	Week 10, Day 1	Week 19
20 Bokeh and Streamlit	Week 10, Day 2	Week 20
21 Capstone Studio and Showcase	Week 11, Day 1	Week 21

🏁 CHECKPOINTS AND CAPSTONE

- **Self-checks and exit tickets** in every session are practice, not grades: they tell you and your instructor what to revisit.
- **Checkpoint 1 (Session 13, 30 minutes):** an individual, open-notebook analysis task on Modules 1 to 4 (NumPy, pandas, charts, statistics), scored with a short rubric.
- **Checkpoint 2 (Session 18, 15 minutes):** a short data-access task on Modules 5 and 6 (SQL and web data).
- **Capstone: The Pinecrest Report.** In Session 20 you choose one of four business questions (why Asheville is falling behind; whether the acquisition channel matters; whether the welcome discount brings people back; what the March 2025 price increase did) and write a one-paragraph plan. Session 21 is a studio with milestone check-ins, then a showcase. You hand in a notebook that reruns from top to bottom, one or two figures, and a one-page report, judged on six criteria: question, data handling, analysis correctness, visuals, communication and reproducibility. An honest, well-supported "no clear difference" scores as well as a dramatic finding.

🧰 REQUIRED SOFTWARE AND DATA

- **Python 3.12 or newer:** with a virtual environment for the course

- **The course packages:** one command: `python -m pip install -r requirements.txt` (NumPy 2.5, pandas 3.0, Matplotlib 3.11, seaborn, SciPy, statsmodels, SQLAlchemy, PyMongo, Requests, Beautiful Soup, Selenium, Plotly, Bokeh, Streamlit)
- **JupyterLab or VS Code:** JupyterLab installs with the packages; VS Code needs its Python and Jupyter extensions
- **Google Chrome:** Selenium drives it in Session 17; Plotly uses it to save charts as images
- **The course data:** `pinecrest-data.zip` and, for working offline in Sessions 16 to 18, `pinecrest-practice-site.zip`, both from the course home page
- **No accounts or paid tools:** the practice API runs on your own laptop, with demo keys given in class

Everything installs once, at home, before Session 1, following Lesson 0.1. Your instructor sends a setup guide and an environment check a week ahead.

✉️ FOR INSTRUCTORS

Want to teach this course? A complete instructor kit (facilitator guides, slides, labs, workbooks, answer keys and assessments for all 21 sessions) is available under license. Ask about the instructor kit.

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